Exact layer margin across the mu-squared neighbourhood with a monotonicity certificate

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

ROBUSTNESS-MU2 remained OPEN because the layer margin m in the off-anchor STEP-5B re-margin was held at its anchor value and only BOUNDED, not recomputed. The closure bar requires the exact $m(\mu^2)$ with $m(\mu^2) \geq 0.4\, m_{\text{anchor}}$ on $[\times 0.5, \times 2]$ at the three certified intensities, the STEP-5B ratio recomputed with the exact margin, and a certified J_{eff} envelope. This note supplies all three plus a monotonicity certificate locating the minimum and a branch-invariance check. Scope: second-cumulant order, $\mu^2 \in [0.0025, 0.01]$, the three certified intensities $\{4 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$.

2. The layer margin is closed-form in mu-squared

$\text{MARGIN}(\mu^2) = P_B(M_+(\mu^2)) - \text{DIP}_{\text{BAND}}(\mu^2)$ with (production convention, bare parameter $R = \mu^2$, gap point $r_R(\mu^2)$, $M_R = M(r_R)$):

$$M_+(\mu^2) = \frac{-3u + \sqrt{9u^2 - 60v\,\mu^2}}{30v}, \qquad \hat{r}_0(M_c) = \mu^2 - \frac{3u^2}{20v}, \qquad \text{DIP}_{\text{BAND}} = \frac{|\hat{r}_0(M_c)|^{3/2}}{3\sqrt{v}},$$

$$P_B(M) = \tfrac{1}{2}(\mu^2 - r_R)(M - M_R) + \tfrac{3}{4}u(M^2 - M_R^2) + \tfrac{5}{2}v(M^3 - M_R^3).$$

This single source (`sectorb_common.margin_of`) reproduces the certified anchor: $\text{MARGIN}(0.005) = 0.004320$.

3. Recomputation and the monotonicity certificate

On a 61-point grid over $[0.0025, 0.01]$:

$$\boxed{\min m(\mu^2) = 0.004082 \text{ at } \mu^2 = 0.0025, \quad m(\mu^2) \geq 0.945\, m_{\text{anchor}}}$$

(the maximum is 0.004802 at $\mu^2 = 0.01$). **Monotonicity certificate (review priority 3).** The central-difference derivative satisfies $\partial_{\mu^2} m(\mu^2) > 0$ at every grid point (min $= 9.5 \times 10^{-2} > 0$), so $m(\mu^2)$ is strictly increasing and the minimum is the LEFT endpoint $\mu^2 = 0.0025$, not an interior point. The structural reason is closed-form: $\text{DIP}_{\text{BAND}}(\mu^2) = |\mu^2 - 3u^2/(20v)|^{3/2}/(3\sqrt{v})$ decreases monotonically ($0.001047 \to 0.000699$ as μ^2 rises toward $3u^2/(20v)$, $\hat{r}_0(M_c)$ becoming less negative), so $-\text{DIP}_{\text{BAND}}$ increases; $P_B(M_+)$ increases over the grid. The $m(\mu^2) \geq 0.4\, m_{\text{anchor}}$ bar is met with factor 0.945.

4. Prop-A branch invariance

Across the whole band, disc $= 9u^2 - 60v\mu^2 > 0$ (minimum 4.7124 at $\mu^2 = 0.01$, well above the two-branch threshold $\mu^2 < 0.0342$) and $M_+ > M_c$: the same upper PD branch is used throughout $[0.0025, 0.01]$, with no branch crossing. The closed-form M_+, P_B, $\mathrm{DIP}_{\mathrm{BAND}}$ are thus valid on the entire interval.

5. STEP-5B ratio (recomputed margin) and the J-eff envelope

μ^2	$m(\mu^2)$	ratio @ 4×10^{-4}	ratio @ 10^{-3}	ratio @ 2×10^{-3}
0.0025 (×0.5)	0.004082	×55.5	×9.16	×2.41
0.005 (anchor)	0.004320	×59.4	×9.80	×2.58
0.01 (×2)	0.004802	×67.5	×11.15	×2.93

The worst ratio over the neighbourhood and the three certified intensities is ×2.41 (the ×0.5, $I = 2\times10^{-3}$ corner) with the EXACT margin. J_{eff} **envelope (review attack 4).** Across the WHOLE 5×3 grid, J_{eff} at quadrature resolutions $n_k = 500$ and $n_k = 1100$ agree to a maximum relative envelope $< 0.01\%$; the worst STEP-5B ratio over the whole grid AND both resolutions is $\times 2.41 > 1$. The verdict is thus J_{eff}-certified everywhere on the grid, not only at the thinnest corner.

6. Closure-bar status

All conditions met: (i) $m(\mu^2)$ recomputed exactly, $\geq 0.945\, m_{\mathrm{anchor}}$, minimum located at $\mu^2 = 0.0025$ by the derivative-sign certificate; (ii) STEP-5B ratio > 1 ($\geq \times 2.41$) with the recomputed margin at the three certified intensities; (iii) J_{eff} envelope certified across the full grid. Remaining qualifier: SCOPE is second-cumulant order. The official status stays **ROBUSTNESS-MU2: OPEN, closure bar met, CLOSE RECOMMENDED**; the flip to CLOSED@[×0.5, ×2]-2ND-CUMULANT is operator-authorized and is NOT performed by this note.

7. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3, 6 code-discipline)

(α) *"exact $m(\mu^2)$ but a 61-point grid."* ADDRESSED: the derivative-sign certificate $\partial_{\mu^2} m > 0$ everywhere plus the closed-form $\mathrm{DIP}_{\mathrm{BAND}}$-decreasing piece prove the minimum is the left endpoint; the grid evaluates that minimum, it does not search for it.

(β) *"J_{eff} certified only at the thinnest corner."* ADDRESSED in §5: the two-resolution envelope is now reported across the whole 5×3 grid ($< 0.01\%$), worst ratio ×2.41.

(γ) *"branch validity on the band."* ADDRESSED in §4: disc > 0 and $M_+ > M_c$ throughout; same PD branch.

(δ) *"this closes the gate."* DISMISSED: the gate stays OPEN; closure is recommended pending operator authorization, second-cumulant scope and [×0.5, ×2] only.

Quantitative sanity check (mandatory): magnitude – MARGIN(0.005) = 0.004320 reproduces the certified 0.00432; sign-direction – $m(\mu^2)$ increasing makes the lower endpoint the worst corner, consistent with the table; limit-case – disc > 0 on the band keeps the two-branch regime; convergence – J_{eff} envelope $< 0.01\%$ over the full grid.

8. Result footer

Result ID:	B1-RH-ENUM / ROBUSTNESS-MU2 exact margin recomputation
Precise statement:	MARGIN(mu^2) = PB(M_+(mu^2)) - DIP_BAND(mu^2), closed-form; reproduces 0.00432 at anchor; over [x0.5,x2] min m = 0.004082 = 0.945 m_anchor at mu^2=0.0025 (proven the minimum by d m/d mu^2 > 0). STEP-5B ratio with the recomputed margin: worst x2.41 > 1 at the three certified intensities. J_eff envelope < 0.01% across the full grid; same PD branch throughout.
Scope:	Second-cumulant order; mu^2 in [0.0025,0.01]; the three certified intensities (not a continuous-intensity claim).
Dependencies:	sectorb_common (margin_of, J_of_t); Prop-A band floor (Math437 v1.2/Math440); production gap_solve/M_fast.
Evidence grade:	EXECUTED (robustness_mu2_margin_recompute.py v1.1, 9/9) + closed-form anchor validation + derivative certificate.
Reproduction command:	python codes/vacuum/robustness_mu2_margin_recompute.py
Expected output:	9/9 PASS; MARGIN(0.005)=0.00432; min m=0.004082 at mu^2=0.0025; d m/d mu^2 > 0; worst ratio x2.41.
Falsification gate:	m(mu^2) < 0.4 m_anchor anywhere; d m/d mu^2 <= 0 somewhere (interior minimum); STEP-5B ratio <= 1.
Tier before / after:	No tier/gate flip. ROBUSTNESS-MU2 closure bar MET; CLOSE@[x0.5,x2]-2ND-CUMULANT RECOMMENDED pending operator sign-off. Gate remains OPEN until then.
No-overclaim statement:	Closure is second-cumulant scope, [x0.5,x2], and the three certified intensities only. The note supplies the exact m(mu^2) + monotonicity certificate; it does not flip the gate. All-orders robustness is SC-SCOPE.
Next required action:	operator authorization to flip ROBUSTNESS-MU2 -> CLOSED@[x0.5,x2]-2ND-CUMULANT (atomic GATES + card + CHANGELOG).